Pre-analysis plan #ManyDaughters

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The aim of this pre-analysis plan is to test the following hypotheses (as causal hypotheses):

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

H2. Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for men respondents with daughters − X for men respondents without daughters) > (X for women respondents with daughters − X for women respondents without daughters)

Hypothesis 1

1) Variables

Which variables will you include? How will these variables be coded— will you use them the way they are or recode them, and if so, how?

Panel data structure: ID (*pid*) & survey year (*syear*)

Dependent variable ($GenderEquality_{it}$): To measure gender equality in terms of attitudes, we use five items with a 7-point response scale. We will construct a mean index across these five items (after reverse-coding where necessary). To assess internal consistency, we will compute Cronbach's alpha to evaluate whether the items form a reliable scale. If the alpha is too low, we will consider excluding one or more items.

Item	Dataset	Variable	Recode
Q467 "Having a job is the best way for a woman to be independent."	pl	plj0621	No (1-7; higher = more equality)
Q468 "Even a married woman should have a paid job so that she can be financially independent."	pl	plj0622	No
Q469 "If a woman earns more money than her partner, this inevitably leads to problems."	pl	plj0623	Reverse values (7-1)
Q470 "For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters."	pl	plj0624	Reverse values
Q471 "At home, the husband should have the final say"	pl	plj0625	Reverse values

Independent variable ($DaughterHH_{it}$): To measure whether respondents have daughters, we use a variable with four categories, based on the relative share of daughters among all children (regardless of the absolute number of children).

Item	Dataset	Variable	Recode
Daughters in HH (*DaughterHH$_{it}$)*	biobirth	kidsex*	0: No children 1: Only son/s or more sons than daughter/s 2: Equal share of son/s and daughter/s 3: More daughters than son/s or only daugther/s

Control variables:

Item	Dataset	Variable	Recode
Gender of respondent (*Woman*$_{it}$)	ppathl	sex	0 = man 1 = woman Various & no answers/ no valid answers = .
Where does your child currently live?	biol	lr3193	1: in household 0: elsewhere
Absolute number of sons and daughters (*NumChildren*$_{it}$)	biol	lb0285	Count variable
Age of children (*ChildAgeGroups*$_{it}$)	biol	age=gebjahr-syear	<1: Infant 1-<4: Toddler 4-<6: Preschooler 6-<13: Schoolage 13-17: Teenager >18: Adult
Socio-economic status, ISEI (*SES*$_{it}$)	pgen bioparen	*pgisei08* *fisei08, misei08*	Metric variable: 0 (low SES) - 100 (high SES)
Partnership status (*Partner*$_{it}$)	ppathl	*partner*	0: no partner in HH 1: in partnership (recode 1, 2, 3, 4, 5)
Sexual orientation (*SexualOrientation*$_{it}$)	ppathl	*sexor*	0: probably heterosexual 1: probably bi/homosexual insufficient information/ no answer = .
Religion (*Religion*$_{it}$)	bioparen	*freli, mreli*	Catholic (1)/ protestant (2)/ other christian denomination (3)/ christian community (7) = Christian; Islamic (4) = Muslim Other denomination (5) = Other; No Denomination (6)/ does not apply (-2) = Non-religious Do not know (0); Else = missings
Age (*Age*$_{it}$)	biobirth	gebjahr	Years as continous variable (syear – gebjahr), no answer = .
East/West (*Region*$_{it}$)	ppathl	sampreg	0: West-Germany 1: East-Germany

2) Inclusion/exclusion criteria

Are there any observations you will exclude? If yes, which? Which years will you include/what period will you study?

Children: We do not restrict the sample by the age of children but only include children who live in the same household.

Unbalanced panel: We make full use of the available observational period (2000 to 2020), requiring a minimum of two years of observation per respondent. The two years do not need to be consecutive. Thus, we work with unbalanced panel data but exclude all individuals with only a single year of observation.

Diversity: We restrict our analysis to respondents who identify as either woman or man, since we expect the number of non-binary/diverse cases to be too small for meaningful comparison. However, we may consider including these groups in robustness checks if sample sizes allow.

Handling of missings: We apply listwise deletion as long as the share of missing values does not reduce the sample size by more than 10%. If missing values exceed this threshold, we will consider alternative strategies—such as using proxy variables (for metric variables) or treating missing values as a separate category (for categorical variables).

3) Exact analysis

Please write down the exact specification of the regression model, including the names of the dependent and independent variables, and whether you use robust standard errors or something else.

To test Hypothesis 1, we estimate a random-effects linear regression model with panel-robust standard errors.

We estimate the following regression model:

$$\text{GenderEquality}_{it} = \beta_0 + \beta_1 \cdot \text{DaughterHH}_{it} + \beta_2 \cdot \text{Woman}_{it} + \beta_3 \cdot \text{Partner}_{it} + \beta_4 \cdot \text{SexualOrientation}_{it} + \beta_5 \cdot \text{Religion}_{it} + \beta_6 \cdot \text{SES}_{it} + \beta_7 \cdot \text{Age}_{it} + \beta_8 \cdot \text{Region}_{it} + \beta_9 \cdot \text{NumChildren}_{it} + \beta_{10} \cdot \text{ChildAgeGroups}_{it} + u_i + \varepsilon_{it}$$

Where:

- i indexes the individual
- t indexes the time (year)
- u_i is the individual-specific random intercept
- ε_{it} is the idiosyncratic error term

4) Interpretation of results

Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test result is in line with the hypothesis or not.

To test Hypothesis 1, we will rely on the regression coefficients and their 95% confidence intervals associated with the independent variable $DaughterHH_{it}$, a categorical measure of child gender composition.

The dependent variable, $GenderEquality_{it}$, captures the attitudinal dimension of gender equality. It is a continuous index constructed as the mean of five survey items on gender role attitudes, reverse-coded where necessary so that higher values indicate stronger support for gender equality.

The reference category will be respondents without children (Category 0). We consider the hypothesis supported if at least one of the following conditions are met:

1. The coefficient for Category 3 (more daughters than sons or only daughters) is positive and statistically significant ($p < 0.05$) compared to Category 0. This would suggest that having daughters is associated with more gender-egalitarian attitudes than being childless.
2. The coefficient for Category 3 is significantly larger than the coefficients for Category 1 (only/more sons) or Category 2 (equal share), indicating that having daughters leads to stronger support for gender equality than having sons or a balanced child gender composition.

If neither condition holds, we will interpret this as no support for the hypothesis.

Hypothesis 2

1) Variables

Panel data structure: ID (*pid*) & survey year (*syear*)

Dependent variable:

Item	Dataset	Variable	Recode
Political attitude ($PoliticalAttitude_{it}$)	pl	plh0004	Reverse values (0=right, 10=left)

Independent and control variables are the same as those used for H1.

2) Inclusion/exclusion criteria

Inclusion and exclusion criteria are the same as those applied in H1.

3) Exact analysis

To test Hypothesis 2, we estimate a random-effects linear regression model with panel-robust standard errors.

We estimate the following regression model:

$$PoliticalAttitude_{it} = \beta_0 + \beta_1 \cdot DaughterHH_{it} + \beta_2 \cdot Female_{it} + \beta_3 \cdot Partner_{it} + \beta_4 \cdot SexualOrientation_{it} + \beta_5 \cdot Religion_{it} + \beta_6 \cdot SES_{it} + \beta_7 \cdot Age_{it} + \beta_8 \cdot Region_{it} + \beta_9 \cdot NumChildren_{it} + \beta_{10} \cdot ChildAgeGroups_{it} + u_i + \varepsilon_{it}$$

Where:

- i indexes the individual
- t indexes the time (year)
- u_i is the individual-specific random intercept
- ε_{it} is the idiosyncratic error term

4) Interpretation of results

To test Hypothesis 2, we will rely on the regression coefficients and their 95% confidence intervals associated with the independent variable $DaughterHH_{it}$, a categorical measure of child gender composition.

The dependent variable, $PoliticalAttitude_{it}$, captures the respondent's ideological self-placement. It is a continuous variable based on an 11-point scale ranging from 0 (right) to 10 (left), reverse-coded so that higher values indicate more left-leaning political attitudes.

The reference category for the independent variable is respondents without children (Category 0). We consider the hypothesis supported if at least one of the following conditions is met:

1. The coefficient for Category 3 (more daughters than sons or only daughters) is positive and statistically significant ($p < 0.05$) compared to Category 0. This would suggest that having daughters is associated with more left-leaning political attitudes than being childless.
2. The coefficient for Category 3 is significantly larger than the coefficients for Category 1 (only/more sons) or Category 2 (equal share), indicating that having daughters leads to more progressive political views than having sons or a balanced child gender composition.

If neither condition holds, we will interpret the results as not supporting the hypothesis.

Hypothesis 3

1) Variables

Panel data structure: ID (*pid*) & survey year (*syear*)

Dependent variable ($GenderEqualityHH_{it}$): To measure gender equality in terms of behavior, we construct a variable based on the relative division of working hours between partners. Specifically, we focus on couples where both partners were interviewed in the same survey year, which allows us to match the respondent's *pid* with their partner's *pid_partner* using the bioparen dataset.

From the individual questionnaires, we extract information on working hours across four domains (e.g., paid work, housework, childcare, and caregiving). For each domain, we calculate the absolute difference in hours between the respondent and their partner. The original scale ranges from negative values (partner works more hours) to positive values (respondent works more hours), with zero indicating equal contribution. To standardize this measure into a gender equality metric, we apply the following transformation:

- Values are recoded such that 0 = full equality,
- ±1 difference becomes 1, ±2 becomes 2, and so on—i.e., $GenderInequality_{it}$ = abs($HoursRespondent_{it}$ – $HoursPartner_{it}$)

This results in a scale from 0 to +∞, where higher values indicate greater inequality in time allocation between partners. Finally, we reverse-code the variable so that higher values correspond to greater behavioral gender equality.

Item	Dataset	Variable	Recode
Q257. Housework, hours per weekday (harmonized)	pl	pli0043_h	No
Q258. Childcare, hours per weekday (harmonized)	pl	pli0044_h	No
Q259. Care for persons in need of assistance, hours per weekday	pl	pli0046	No
Q260. Repairs/gardening, hours per weekday (harmonized)	pl	pli0049_h	No

Independent and control variables are the same as those used for H1.

2) Inclusion/exclusion criteria

Inclusion and exclusion criteria are the same as those applied in H1.

3) Exact analysis

To test Hypothesis 3, we estimate a random-effects linear regression model with panel-robust standard errors.

We estimate the following regression model:

$$GenderEqualityHH_{it} = \beta_0 + \beta_1 \cdot DaughterHH_{it} + \beta_2 \cdot Female_{it} + \beta_3 \cdot Partner_{it} + \beta_4 \cdot SexualOrientation_{it} + \beta_5 \cdot Religion_{it} + \beta_6 \cdot SES_{it} + \beta_7 \cdot Age_{it} + \beta_8 \cdot Region_{it} + \beta_9 \cdot NumChildren_{it} + \beta_{10} \cdot ChildAgeGroups_{it} + u_i + \varepsilon_{it}$$

Where:

- i indexes the individual
- t indexes the time (year)
- u_i is the individual-specific random intercept
- ε_{it} is the idiosyncratic error term

For robustness checks, we repeat the regression analysis separately for each individual item of the dependent variable.

4) Interpretation of results

To test Hypothesis 3, we will rely on the regression coefficients and their 95% confidence intervals associated with the independent variable $DaughterHH_{it}$, a categorical measure of child gender composition.

The dependent variable, $GenderEqualityHH_{it}$, captures the behavioral dimension of gender equality. The final variable is continuous and ranges from 0 (maximum inequality) to a rescaled maximum value, where higher values indicate greater equality in the division of labor between partners.

The reference category for the independent variable is respondents without children (Category 0). We consider the hypothesis supported if at least one of the following conditions is met:

1. The coefficient for Category 3 (more daughters than sons or only daughters) is positive and statistically significant ($p < 0.05$) compared to Category 0. This would suggest that having daughters is associated with a more equal division of labor between partners.
2. The coefficient for Category 3 is significantly larger than the coefficients for Category 1 (only/more sons) or Category 2 (equal share), indicating that having daughters promotes greater gender equality in household labor than having sons or a balanced child gender composition.

If neither condition holds, we will interpret the result as not supporting the hypothesis.

Hypothesis 4

1) Variables

Panel data structure: ID (*pid*) & survey year (*syear*)

Independent variable:

Item	Dataset	Variable	Recode
Gender of respondent (Man_{it})	ppathl	sex	1 = man 0 = woman
Daughters in HH *(*$DaughterHH_{it}$*)*	biobirth	kidsex*	0: No children 1: Only son/s or more sons than daughter/s 2: Equal share of son/s and daughter/s 3: More daughters than son/s or only daugther/s

Dependent and control variables are the same as for H1, except that the variable $Woman_{it}$, previously used as a control, is now the independent variable, recoded as Man_{it}.

2) Inclusion/exclusion criteria

Inclusion and exclusion criteria are the same as those applied in H1.

3) Exact analysis

To test Hypothesis 4, we estimate a random-effects linear regression model with panel-robust standard errors.

$$GenderEquality_{it} = \beta_0 + \beta_1 \cdot DaughterHH_{it} + \beta_2 \cdot Male_{it} + \beta_3 \cdot (DaughterHH \times Man)_{it} + \beta_4 \cdot Partner_{it} + \beta_5 \cdot SexualOrientation_{it} + \beta_6 \cdot Religion_{it} + \beta_7 \cdot SES_{it} + \beta_8 \cdot Age_{it} + \beta_9 \cdot Region_{it} + \beta_{10} \cdot NumChildren_{it} + \beta_{11} \cdot ChildAgeGroups_{it} + u_i + \varepsilon_{it}$$

Where:

- i indexes the individual
- t indexes the time (survey year)
- u_i is the individual-specific random intercept
- ε_{it} is the idiosyncratic error term

4) Interpretation of results

To test Hypothesis 4, we will rely on the regression coefficients and their 95% confidence intervals associated with the independent variable $DaughterHH_{it}$, the binary gender of the respondent (Man_{it}), and their interaction term ($DaughterHH \times Man_{it}$).

The dependent variable, $GenderEquality_{it}$**,** captures the attitudinal dimension of gender equality. It is a continuous index constructed as the mean of five survey items on gender role attitudes, reverse-coded where necessary so that higher values indicate stronger support for gender equality.

We consider the hypothesis supported if the interaction term ($DaughterHH \times Man$) is statistically significant and positive ($p < 0.05$). This would indicate that the effect of having daughters on gender-egalitarian attitudes is stronger for men than for women, as hypothesized.

If the interaction term is not statistically significant or points in the opposite direction, we will interpret this as no support for the hypothesis.